Nykko Vitali

Department of Psychology, Harvard University

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Education

2024–present **Ph.D. Student in Psychology**, Harvard University

2018–2020 **B.A. in Psychology**, *cum laude*, University of Pennsylvania

2016–2018 **A.A. in Psychology**, *with highest honors*, Community College of Philadelphia

Research Interests

Human–AI interaction, machine psychology, social cognition, metacognition, natural language processing

Publications

Preprints

Vitali, N., & Banaji, M. R. (2026). *Large language models exhibit human-like Bayesian hypocrisy* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2609.35779

Journal Articles

Lehr, S. A., Saichandran, K. S., Harmon-Jones, E., **Vitali, N.**, & Banaji, M. R. (2025). Kernels of selfhood: GPT-4o shows humanlike patterns of cognitive dissonance moderated by free choice. *Proceedings of the National Academy of Sciences*, *122*(20), e2501823122. https://doi.org/10.1073/pnas.2501823122

Rhodes, E., Mechanic-Hamilton, D., Phillips, J. S., Bahena, A., **Vitali, N.**, Hlava, Q., Cook, P., Gee, J., Grossman, M., McMillan, C., & Massimo, L. (2024). Discrepancies in patient and caregiver ratings of personality change in Alzheimer's disease and related dementias. *Archives of Clinical Neuropsychology*, *39*(4), 399–408. https://doi.org/10.1093/arclin/acad079

Comments and Replies

Lehr, S. A., Saichandran, K. S., Harmon-Jones, E., **Vitali, N.**, & Banaji, M. R. (2025). Reply to Cummins et al.: GPT reveals cognitive dissonance that is both irrational and alarmingly humanlike. *Proceedings of the National Academy of Sciences*, *122*(35), e2518613122. https://doi.org/10.1073/pnas.2518613122

Poster Presentations

2026 **Sustained Conversation Breeds Suspicion, Not Accuracy, in AI Detection**
Poster presented at the AI-SDM Workshop on Human-AI Complementarity for Decision Making, Carnegie Mellon University.

2026 **Distinguishing Human From AI: A Process-Level Exploration**
Poster presented at the Diverse Intelligences Summit, Princeton University; and the Symposium on Communication & Intelligence, University of Chicago Data Science Institute.

2026 **Large Language Models Exhibit Human-Like Bayesian Hypocrisy**
Poster presented at the Society for Personality and Social Psychology (SPSP) Annual Convention.

Media Coverage

2026 **Tech Xplore**, coverage of Vitali & Banaji (2026), with interview

2025 **Harvard Gazette**, **PsyPost**, **Neuroscience News**, coverage of Lehr et al. (2025)

Awards, Grants, and Research Support

2026 **Pangram Labs API Research Credits**, in-kind research support, $5,000

2026 **GPTZero API Access**, in-kind research support

2025 **Rand Fund Grant**, Using NLP to Study Online Moral Discourse, $8,000

2018 **Magner Nichols Public Speaking Award**, 2nd Place

2016 **Phi Theta Kappa Honor Society**

Teaching and Mentoring

Fall 2026 **Teaching Fellow**, Gen Ed 1201: *Foundations of NeuroAI*, Harvard University, Cambridge, MA
Teaching two sections of a General Education course on the overlap between cognitive neuroscience and machine learning. Lead weekly sections, grade coding-based problem sets, and hold office hours. Instructor: Prof. Talia Konkle.

June–July 2026 **Undergraduate Research Mentor**, Harvard University, Cambridge, MA
Supervised one undergraduate applying natural language processing methods to Turing-test transcript data. Co-developed analyses of which conversational cues judges relied on when classifying human versus AI interlocutors, and which of those cues were actually diagnostic. Trained the student in text preprocessing, feature extraction, and interpretation of model output.

July 2026 **Research Mentor**, Summit Institute, Summer Research Program at Harvard University, Cambridge, MA
Mentored three high school students one-on-one through a two-week intensive research program. Each student produced an original research paper as the goal of participation.

Research and Professional Experience

2022–2024 **UX Research Assistant**, Exponent

2021–2022 **Clinical Research Coordinator**, Frontotemporal Degeneration Center, University of Pennsylvania

Technical Projects

2026 **Real-Time Interaction Study Platform** *GitHub repository*
Built a multiplayer web app for human-AI distinction studies using Python (FastAPI) and reactive JavaScript. Includes robust matchmaking, dynamic LLM persona injection (Warm, Contrarian, Neutral), and dropout-resilient session handling. Logs high-dimensional behavioral data, including continuous confidence and keystroke dynamics, to PostgreSQL.

2025 **ClaudeR** *GitHub repository*
Developed an R package that bridges RStudio with LLMs, enabling interactive coding sessions where LLMs execute code, iteratively solve errors, and display results in real time.

2024 **Neuroimaging AR Pipeline for Vision Pro** *LinkedIn post*
Developed a processing pipeline to convert 7T MRI scans into interactive 3D brain models for Apple Vision Pro using FreeSurfer, custom Python scripts, and Blender for anatomical visualization.

Technical Skills

Programming R, Quarto, Python, Swift, Bash, MCP

Databases PostgreSQL, MongoDB

Web HTML, CSS, JavaScript, Flask